Updated 15 October 2026

M. Colin Marvin

Postdoctoral Research Associate, Brown University

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APPOINTMENTS

2026-present	Postdoctoral Research Associate	Brown University

Advisor: Prof. Samuel P.D. Birch

EDUCATION

2021-2026	Ph.D.	Geological Sciences	Stanford University

Advisor: Prof. Mathieu G.A. Lapôtre
Thesis: *"Patterns in grains and landscapes as environmental records"*

2017-2021	B.S.	Geography (Honors)	Arizona State University
2017-2021	Minor	Mathematics	Arizona State University
2017-2021	Certificate	GIS	Arizona State University

Advisor: Prof. Ian J. Walker
Thesis: *"Disturbance signatures and sediment characteristics in aeolian dune landscapes"*

AWARDS AND HONORS

2026	**Outstanding Graduate Paper Award.** *International Society for Aeolian Research.*
2025	**Outstanding Student Presentation Award (OSPA).** *American Geophysical Union*.
2021	**Dean's Medal.** *School of Geographical Sciences and Urban Planning, Arizona State University*.
2019	'**The College' Student Leader.** *School of Geographical Sciences and Urban Planning, Arizona State University*.

GRANTS AND FELLOWSHIPS

2023	**McGee-Levorsen Fellowship.** ($2,300).
2022	**NCALM Seed Grant.**
2021	**Geological Sciences Travel Fund.** ($500).
2020	**Barrett Global Explorers Grant**. ($5,000).

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INVITED TALKS, SEMINARS, AND LECTURES

2026 **Planetary Lunch Seminar.** Department of Earth, Atmospheric, and Planetary Sciences, *Massachusetts Institute of Technology*.
2026 **Planetary Lunch Seminar.** Department of Earth, Environmental, and Planetary Sciences, *Brown University*.
2026 **Seminar.** *International Sand Collectors Society*.
2026 **Department Seminar.** Department of Space Sciences, *Southwest Research Institute (Boulder).*
2025 **Virtuaeolian Seminar Series.** *International Society for Aeolian Research*.
2025 **Department Seminar.** Department of Environmental Studies and Sciences, *Santa Clara University.*
2024 **Planetary Science and Exploration Seminar.** Department of Geophysics, *Stanford University.*

PEER-REVIEWED PUBLICATIONS

In review, in revision, or in prep

A. **Marvin, M.C.**, Gleason, A., Narteau, C., Rozier, O., & Lapôtre, M.G.A. (in review). Windblown dunes are born in sand at the brink of transport.
B. Blaske, C., **Marvin, M.C.**, & Lapôtre, M.G.A. (in review). Megaflood duration estimates from bedform pattern disequilibrium.
C. Blaske, C.H., **Marvin, M.C.**, & Lapôtre, M.G.A. (in prep). Duration of outflow channel-carving floods on Mars.
D. **Marvin, M.C.** & Lapôtre, M.G.A. (in prep). Surface wind conditions on Mars from bedform morphodynamics.
E. Colicci, V., **Marvin, M.C.**, Hasson, M., Altermann, W., & Lapôtre, M.G.A. (in prep). Earth's Oldest Known Coastal Aeolian Dune Field: Sedimentology and Stratigraphy of a Vryburg Formation Quartzite, Transvaal Supergroup, South Africa
F. Colicci, V., **Marvin, M.C.**, Hasson, M., Altermann, W., & Lapôtre, M.G.A. (in prep). Atmospheric density in the Neoarchean from aeolian bedform dimensions.
G. Radebaugh, J., Wright, M., Rose, D., **Marvin, M.C.**, Lapôtre, M.G.A., & Gunn, A. (in prep). A Global Compendium of Dune Long Axis Traces on Titan Reveals Wind and Topographic Controls.

Published

10. Hasson, M., **Marvin, M.C.**, & Lapôtre, M.G.A. (2026). Zircon transport histories revealed by automated classification of scanning electron microscope images. *Journal of Sedimentary Research*. https://doi.org/10.2110/jsr.2026.050
9. **Marvin, M.C.**, Hasson, M., Colicci, V., Abubo, R., & Lapôtre, M.G.A. (2026). Microtextural analyses of detrital zircon for paleoenvironmental interpretations of metasedimentary rocks. *Geology*, *54*(1), 35-39. https://doi.org/10.1130/G53712.1
8. **Marvin, M.C.**, Lapôtre, M.G.A., Radebaugh, J., & Bo, W. (2025). From Xanadu around and back: A ca. 11,000 km journey of windblown sand revealed by global dune patterns on Titan. *Geophysical Research Letters*, 52, e2024GL112760. https://doi.org/10.1029/2024GL112760
7. Hasson, M., **Marvin, M.C.**, & Lapôtre, M.G.A. (2024). Automated determination of transport and depositional environments in sand and sandstones. *Proceedings of the National*

Academy of Sciences of the United States of America, *121*(40), e2407655121. https://doi.org/10.1073/pnas.2407655121

6. **Marvin, M.C.**, Lapôtre, M.G.A., Gunn, A., Day, M.D., & Soto, A. (2023). Dune interactions record changes in boundary conditions, *Geology*, *51*(10), 947-951. https://doi.org/10.1130/G51264.1
5. Hasson, M., **Marvin, M.C.**, Gunn, A., Ielpi, A. & Lapôtre, M.G.A. (2023). A depositional model for meandering rivers without land plants. *Sedimentology*, *70*(7), 2272-2301. https://doi.org/10.1111/sed.13121
4. Heintzman, R., Brandi, A., Kelley, M., & **Marvin, M.C.** (2023). A Physical Geography Lab's Online Transition: Student and Instructor Insights Using iGEO Video Games during the Pandemic. *Journal of Geography,* 1-9. https://doi.org/10.1080/00221341.2023.2216705
3. González, C., Kelley, M., **Marvin, M.C.**, López-Castañeda, N., Dorn, R.I., & Schmeeckle, M. (2022). Regional piedmont incision during base-level rise in the northeastern Sonoran Desert, Arizona, USA. *Physical Geography*, *43*(1), 67-97. https://doi.org/10.1080/02723646.2021.1934964
2. Shandonay, K.L., Moll, H., **Marvin, M.C.**, López-Castañeda, N., Kelley, M., Hilgendorf, Z., Heintzman, R., & Dorn, R. (2021). The Fieldwork of Shared Experiences. *The Geographical Bulletin,* *62*(2), 82-88. https://www.gammathetaupsilon.org/the-geographical-bulletin/2020s/volume62-2/A/article7.pdf
1. Hilgendorf, Z., **Marvin, M.C.**, Turner, C.M., & Walker, I.J. (2021). Assessing Geomorphic Change in Restored Coastal Dune Ecosystems Using a Multi-Platform Aerial Approach. *Remote Sensing*, *13*(3), 354. MDPI AG. https://doi.org/10.3390/rs13030354

NON-PEER-REVIEWED PUBLICATIONS

b. **Marvin, M.C.**, Abbey, P., Mrazek, J., Weitzel, E., & Work, R. (2021). Assateague Island National Seashore Ecological Forecasting: Characterizing Nearshore Suspended Sediments and Landcover Change Relative to Sediment Bypassing and Catastrophic Events. NASA DEVELOP National Program.

a. Dialesandro, J., Kruskopf, M., **Marvin, M.C.**, & Vargas, M. (2021). San Diego Urban Development: Utilizing NASA Earth Observations to Identify Drivers of Extreme Urban Heat and Generate a High-Resolution Vulnerability Index for Urban Planning and Climate Resiliency in San Diego, California. NASA DEVELOP National Program.

CONFERENCE PRESENTATIONS

First-authored:

Marvin, M.C., Fernandez, B., & Birch, S.P.D. (2026). Deciphering the Enigmatic Record of Titan's Climate from River-Dune Interactions. Submitted to *AGU Fall Meeting 2026*.

Marvin, M.C., Gleason, A., Lapôtre, M.G.A., Narteau, C., & Rozier, O. (2026). Windblown dunes are born in sand at the brink of transport. Submitted to *AGU Fall Meeting 2026*.

Marvin, M.C. & Lapôtre, M.G.A. (2026). Surface wind conditions on Mars from bedform morphodynamics. Submitted to *Boston Area Planetary Science Meeting 2026*.

Marvin, M.C., Gleason, A., Lapôtre, M.G.A., Narteau, C., Rozier, O., & Alkalla, J. (2026). Do Aeolian Dune Fields Evolve to a Near-Critical Threshold State? *International Conference on Aeolian Research XII*.

Marvin, M.C. & Lapôtre, M.G.A. (2025). Do Aeolian Dune Fields Evolve to a Near-Critical Threshold State? *AGU Fall Meeting 2025*, EP31D-1678.

Marvin, M.C., Lapôtre, M.G.A., Radebaugh, J., & Bo, W. (2025). From Xanadu around and back: A ca. 11,000 km journey of windblown sand revealed by global dune patterns on Titan. *8th International Planetary Dunes Conference,* Abstract #3005.

Marvin, M.C., Hasson, M., Abubo, R., & Lapôtre, M.G.A. (2024). Zircon Microtextures: A Record of Earth's Earliest Surface Environments. *AGU Fall Meeting 2024*, EP43F–03.

(*invited*) **Marvin, M.C.**, Lapôtre, M.G.A., Radebaugh, J., Bo, W., Day, M., Gunn, A., & Soto, A. (2024). Surface conditions on Titan through dune-pattern analysis. *GSA 2024*. Abstract #401270.

Marvin, M.C., Hasson, M., Abubo, R., & Lapôtre, M.G.A. (2024). Detrital zircon microtextures as a powerful tool to interpret the Precambrian sedimentary record. *International Sedimentary Geoscience Congress 2024*, Abstract #80.

Marvin, M.C., Bo, W., Radebaugh, J., Gunn, A., Day, M.D., & Lapôtre, M.G.A. (2023). Global analysis of dune patterns on Titan. *AGU Fall Meeting 2023*, EP31D–2112.

Marvin, M.C., Lapôtre, M.G.A., Gunn, A., Day, M., & Soto, A. (2023). Dune interactions as an indicator of morphodynamic disequilibrium. *International Conference on Aeolian Research XI,* Abstract #324.

Marvin, M.C., Bo, W., & Lapôtre, M.G.A. (2023). What can and cannot be learned from dune interactions on Titan from Cassini SAR images. *Titan Through Time VI.*

Marvin, M.C., Gunn, A., Day, M., & Lapôtre, M.G.A. (2022). Quantifying dune interactions on planetary surfaces: exploring pattern development dependence on environmental conditions. *7th International Planetary Dunes Conference,* Abstract #3023.

Marvin, M.C., Gunn, A., Day, M., & Lapôtre, M.G.A. (2022). Quantifying dune interactions on planetary surfaces: updated methodology and implications for dune pattern analyses. *53rd Lunar and Planetary Science Conference,* Abstract #1236.

Marvin, M.C. (2021). Effectiveness of foredune restoration in high-energy coastal environments. *AAAS: Understanding Dynamic Ecosystems.* (2nd Place, Environment and Ecology).

Contributing author:

Fernandez, B., Birch, S.P.D., Umurhan, O., **Marvin, M.C.**, Mustard, J.F., & Radebaugh, J. (2026). Modeling Coupled Fluvial-Aeolian Dynamics at the Kuiseb River, Namibia, with Implications for Titan's Xanadu Margin. Submitted to *AGU Fall Meeting 2026*.

Blaske, C., **Marvin, M.C.**, & Lapôtre, M.G.A. (2026). Duration of outflow-channel forming floods from bedform pattern dynamics at Athabasca Valles, Mars. Submitted to *AGU Fall Meeting 2026*.

Kevorkian, M., Looker, N., **Marvin, M.C.**, Rahalski, E., Laan, M., Homyak, P.M., Lapôtre, M.G.A., & Willenbring, J.K. (2026). From Bedrock to Soil: Grain Microtextures Track Soil Weathering and Residence Time on Hillslopes. Submitted to *AGU Fall Meeting 2026*.

Colicci, V., **Marvin, M.C.**, Hasson, M., Altermann, W., & Lapôtre, M.G.A. (2026). Sedimentology and stratigraphy of a Neoarchean coastal dune field. *International Conference on Aeolian Research XII*.

Umurhan O. M., Magana L. O., **Marvin, M.C.**, Fernandez B., Radebaugh, J., Soderblom, J.M., Kaplan, S., Birch, S.P.D., & Moore, J.M. (2026). MARSSIM's New Sand Dune Modeling Capability for Planetary Science Applications: From the Namib Desert to Io and Titan. *57th Lunar and Planetary Science Conference,* Abstract #1371.

Colicci, V., **Marvin, M.C.**, Hasson, M., Altermann, W., & Lapôtre, M.G.A. (2025). Sedimentology and stratigraphy of one of Earth's Oldest reported dune fields: the 2.64 Ga Vryburg Formation of the Transvaal Supergroup, Griqualand West, South Africa. *AGU Fall Meeting 2025*, EP21E-1659.

Blaske, C., **Marvin, M.C.**, & Lapôtre, M.G.A. (2025). Outburst flood duration on Earth and Mars from bedform pattern disequilibrium. *AGU Fall Meeting 2025*, EP34A-02.

Lester, H., Kelley, M., Sankey, J., Caster, J., Byerley, E., Fairley, H., and **Marvin, M.C.** (2025). Evaluating ERA5-Land wind data across plateau and canyon environments in the Grand Canyon region. *AGU Fall Meeting 2025*, EP31D-1680.

Lui, T., **Marvin, M.C.**, Caers, J., & Lapôtre, M.G.A. (2024). Segmentation of Dune Crestlines Using Convolutional Neural Networks. *AGU Fall Meeting 2024*, EP51D–1378.

Radebaugh, J., Wright, M., Rose, D., Kerber, L., Lapôtre, M.G.A., Rodriguez, S., Lorenz, R.D., Barnes, J.W., **Marvin, M.C.**, & Cohen-Zada, A. (2024). Eolian landforms and processes on Titan and role in material movement. *Europlanet Science Congress 2024*, EPSC2024-613.

Hasson, M., **Marvin, M.C.**, & Lapôtre, M.G.A. (2024). Determining depositional environments using quartz microtextures and deep learning. *International Sedimentary Geoscience Congress 2024*, Abstract #174.

Hasson, M., **Marvin, M.C.**, & Lapôtre, M.G.A. (2023). Determination of paleo-transport environments of sand grains using deep learning. *AGU Fall Meeting 2023*, EP31D–2112.

Hasson, M., **Marvin, M.C.**, Gunn, A., Ielpi, A., & Lapôtre, M.G.A. (2023). Preservation of meandering river deposits in unvegetated arid landscapes: Implications for paleoenvironmental interpretations of fluvial deposits on the pre-vegetation Earth and Mars. *International Conference on Fluvial Sedimentology 2023*, Abstract S03–04.

Hasson, M., **Marvin, M.C.**, & Lapôtre, M.G.A. (2022). A depositional model for meandering rivers in unvegetated arid basins. *AGU Fall Meeting 2022*, EP56A–06.

Heintzman, R., Brandi, A., Kelley M.M., **Marvin, M.C.** (2021). Student and Instructor Insights for an Online Synchronous Introductory Geography Lab Using Interactive Geovisualization "Video Games" – Fall 2020 Term at Arizona State University. *American Association of Geographers (AAG) Annual Meeting.*

TEACHING

2026 **EEPS 0070—Introduction to Oceanography (Guest Lectures).** *Department of Earth, Environmental, and Planetary Sciences, Brown University*

2026 **EEPS 0080—Art in Flight: The Creative Exploration of Titan (Guest Lecture).** *Department of Earth, Environmental, and Planetary Sciences, Brown University*

2023 **GEOLSCI 120/220—Planetary Surface Processes: Shaping the Landscape of the Solar System (TA).** *Department of Earth and Planetary Sciences, Stanford University*.

2022 **GEOLSCI 180/280—Magmatic and Eruptive Processes (TA).** *Department of Earth and Planetary Sciences, Stanford University*.

2018–2021 **GPH 112—Physical Geography Lab (Peer Mentor).** *School of Geographical Sciences and Urban Planning, Arizona State University*.

SERVICE AND COMMUNITY ENGAGEMENT

Ongoing **Peer Reviewer** for *Nature Communications, Geology, Geophysical Research Letters, Journal of Geophysical Research: Earth Surface, Journal of Geophysical Research: Planets, Earth and Planetary Science Letters, Icarus, Aeolian Research,* and *The Geographical Bulletin.*

2026-present **Letters to a Pre-Scientist**. Started January 2026.

2019-present **Member: Gamma Theta Upsilon.** National Honor Society in Geography. Since March 2019.

2026 **Panel Reviewer** for *NSF GEO.*

2022-2026 **Graduate Student Mentor**. *Department of Earth and Planetary Sciences, Stanford University*.
2021-2026 **Graduate Student Representative, Graduate Studies Committee**. *Department of Earth and Planetary Sciences, Stanford University*.
2025 **Peer Mentor.** *American Geophysical Union Earth and Planetary Surface Processes*.
2024-2025 **Session Convener** at the *American Geophysical Union Fall Meeting* (2024, 2025).
2024-2025 **GeoKids**. *Doerr School of Sustainability, Stanford University*.
2024 **Eclipse Outreach**. *Department of Earth and Planetary Sciences, Stanford University*.
2023 **SESUR Mentor**. *Department of Earth and Planetary Sciences, Stanford University*.

OTHER PROFESSIONAL AND RESEARCH EXPERIENCE

2021 **Project Lead, NASA DEVELOP Program,** *Science Systems and Applications (SSAI)*.
2021 **Summer Undergraduate Research Intern,** *Arizona State University* (Advisors: Prof. R. Muenich, Dr. D. Schaffer-Smith, Dr. J. DeMeester).
2021 **Team Member, NASA DEVELOP Program,** *Science Systems and Applications (SSAI)*.
2020 **Content Curation Fellow,** *SolarSPELL.*
2019 **Urban Equity Initiative Fellow.** *School of Geographical Sciences and Urban Planning, Arizona State University.*
2018-2021 **Research Assistant,** *Arizona State University*. (Advisor: Prof. I. Walker)

FIELD EXPERIENCE

2025 Namib Sand Sea, Namibia (drone survey); Channeled Scablands, Washington (drone survey); South Africa (stratigraphy/sedimentology, drone survey); Death Valley National Park (field trip)
2024 South Africa (stratigraphy/sedimentology); Dixie Valley, Nevada (drone survey)
2023 Grand Staircase Escalante National Monument (drone survey; rock sampling); Mississippi River (sediment sampling)
2022 Dixie Valley, Nevada (stratigraphy/sedimentology, drone survey); Clear Lake, California (field trip); Lassen Volcanic National Park (field trip)
2018 Joshua Tree National Park (drone survey); Lanphere Dunes National Natural Landmark (drone/lidar survey)

TECHNICAL SKILLS

Software Python, MATLAB, ArcPro, QGIS, Google Earth Engine
Hardware RTK-GPS, drones, scanning electron microscope, rock crushing and mineral separations

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MEDIA MENTIONS

2026 "A monster flood hit Death Valley. Dune recovery is erasing the evidence": Las Vegas Review-Journal

2025 "Colin Marvin - Zircon microtextures: A record of Earth's earliest surface environments - Virtuaeolian": *International Society for Aeolian Research*

2024 "Automated determination of transport and depositional environments in sand and sandstones": *NPR Science Friday*, *MSN*, *Stanford Report*, *Interesting Engineering*

2024 Eclipse Outreach 2024: *The Stanford Daily*, *CBS News*

2023 "Dune interactions record changes in boundary conditions": *Stanford Earth*, *Phys.org*, *Science Daily*

2022 "NCALM Announces 2021 Seed Proposal Winners": *NCALM*

2021 "In pursuit of global impact, geography Dean's Medalist to study planetary surface processes of Saturn moon": *ASU News*

2020 "Barrett students win Barrett Global Explorers Grant for worldwide research projects": *Barrett, The Honors College*

2019 "New ASU fellowship program addresses urban equity through geographic perspectives": *ASU News*